

Hand Gesture Recognition with Deep Learning and Transfer Learning

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Abstract

Hand gesture recognition in the wild always faces two primary challenges: cluttered backgrounds and complex spatial-temporal dependencies. For American Sign Language (ASL), the meaning is often embedded in the precise configuration of fingers. This project proposes a robust two-stage pipeline to translate ASL video into English words, trying to provide a solution for real-time sign language interpretation.

1. Introduction

1.1. Mathematical Foundation

A standard MLP treats each input pixel as an independent feature, applying a fully-connected weight matrix across the entire input. For an image of size $H \times W \times C$, this means the first layer alone requires $H \times W \times C \times N$ parameters, which grows prohibitively with resolution and captures no spatial structure. More critically, MLPs have no inductive bias for locality — a pixel in the top-left corner is treated as equally related to its immediate neighbor as to a pixel on the opposite side of the image.

CNNs address this by replacing global matrix multiplication with localized convolution operations. A convolutional kernel of size $k \times k$ shares weights across all spatial positions, reducing the parameter count from $O(H \times W \times N)$ to $O(K^2 \times N)$. This locality is formalized through the concept of the receptive field: each neuron in a convolution layer only responds to a $K \times K$ region of the input. By stacking multiple layers, the effective receptive field grows geometrically, allowing deeper layers to capture progressively larger structural patterns without ever requiring explicit global connections.

However, CNNs are fundamentally constrained by this same locality, while Vision-Transformers (ViT) eliminate this constraint entirely by dividing the image into fixed-size patches and treating as a token in a self-attention sequence. The attention mechanism computes pairwise relationships between all patches simultaneously, giving every position a global receptive field from the very first layer, which allows ViT to capture dependencies that CNNs struggle with.

1.2. Validation on CNN

In this part, we do the validation by constructing a CNN-based classifier from scratch using PyTorch. The network was based on ResNet-18, with 3×3 kernel instead of 7×7 in Stem, which have better performance on small sized images.

	Accuracy	Parameters
CIFAR-10	0.795	2,777,674
CIFAR-10	0.513	676,682
Sign MNIST	0.947	-
Sign MNIST	0.997	677,465

Table 1. Validation on CNN

We evaluated two ResNet variants, a 3-conv stages model and a shallower model (2 conv stages) on CIFAR-10 and Sign MNIST respectively. The layer3 have 256 in_channels and 256 out_channels.

On CIFAR-10, removing layer3 caused a significant accuracy drop from 79.5% to 51.3%. CIFAR-10 contains 10 semantically diverse categories (animals, vehicles, etc.) with complex textures and inter-class variation, requiring deeper feature hierarchies to discriminate. The shallower model lacks the representational capacity to capture these high-level abstractions, leading to underfitting.

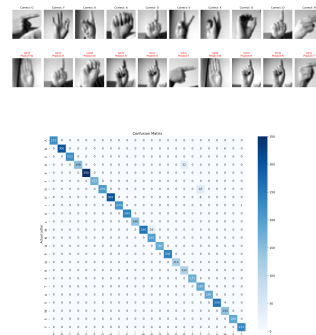


Figure 1. Prediction on Sign MNIST

On Sign MNIST, however, the shallower model outper-

formed the deeper one, Sign MNIST consists of 28×28 grayscale images of hand gestures with relatively low intra-class variation and simple spatial structure. In this case, the additional capacity of layer3 provides no benefit and instead introduces a higher risk of overfitting given the dataset’s limited complexity. The shallower model generalizes better precisely because the task does not demand deep feature extraction. In Figure 1, we attached some predictions and confusion matrix.

1.3. Validation on ViT

The idea of how to validate the ViT is initially from [Sanford CS231n](#). We constructed a ViT from the scratch rely on PyTorch and applied a 2-stage validation: an Encoder-Decoder Transformer for image captioning, and a Decoder-only ViT for image classification.

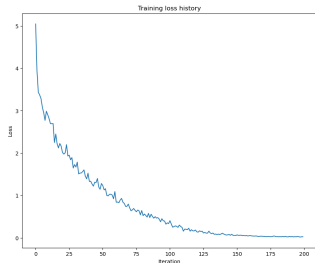


Figure 2. Loss for Image Captioning

In the first phase, we constructed an Encoder-Decoder Transformer, and run for 100 epochs, the last epoch loss dropped to 0.026, which is lower than the target (0.05).

In the second phase, we applied this ViT on both CIFAR-10 and Sign MNIST as we did in Section 1.2, with 2-epoch training on whole training dataset, and achieved the following:

	Accuracy	Baseline	Parameters
CIFAR-10	0.466	0.45	802,442
Sign MNIST	0.939	—	1,592,089

Table 2. Validation on ViT

Our ViT achieved 46.6% on CIFAR-10 and 93.9% on Sign MNIST, both under-performing compared to the CNN-based ResNet we did in Section 1. This is consistent with well-established findings in the literature: ViT lacks the inductive biases inherent to CNN (locality, translation equivalence), making it data-hungry and poorly suited for small-scale datasets without pretraining.

From Table 1 and 2, we can observe that on CIFAR-10, the extremely short token sequence (only 16 patches at patch size 8) further limits the model’s ability to leverage global attention, resulting in performance barely above the

trivial baseline. On Sign MNIST, the simpler visual structure allows ViT to approach CNN-level performance, but the parameter overhead (1.59M vs. 677K) yields diminishing returns.

2. Transfer Learning

2.1. Architectural Benchmarking

To evaluate the representational quality of different video backbones on the WLASL-300 sign language recognition task, we conducted linear probe experiments across four architectures: S3D (CNN-based), VideoMAE (Vision Transformer-based), VideoMAE pretrained on a different dataset, and an LSTM-based baseline. In all linear probe settings, the backbone weights are fully frozen and only a lightweight classification head is trained, isolating the quality of the pretrained features from the effect of task-specific optimization. Results are summarized in the table below:

Method	Train Acc	Test Acc	Latency (ms)	FPS
S3D	32.79%	8.30%	2.25	445.35
VAE	48.97%	8.52%	1.48	674.8

Table 3. Linear probe results on S3D and VAE.

First, all linear probe models exhibit a pronounced gap between training and test accuracy, with test performance ranging from approximately 8% to 20% on a 300-class task (random chance is about 0.33%). This indicates that while the frozen backbones capture some transferable features, none are sufficiently aligned with the fine-grained hand gesture patterns required for sign language recognition to support strong generalization under linear evaluation alone.

Method	Test Acc	Pretrained Dataset
VAE	8.52%	Kinetics-400
VAE	20.18%	SSv2

Table 4. Effect of pretraining dataset on VAE.

Second, the choice of pretraining data has a larger impact than architecture. Both S3D and VideoMAE pretrained on Kinetics-400 achieve similar test accuracy (8.5%), despite their architectural differences. Kinetics-400 contains coarse-grained, full-body action categories that share little visual similarity with isolated hand signs. In contrast, switching VideoMAE’s pretraining to Something-Something V2 (SSv2), test accuracy is adoubled to 20.18%, while maintaining identical architecture and training procedure. This directly demonstrates that pretraining domain alignment is a stronger determinant of transfer quality than model architecture in this setting.

Third, VideoMAE consistently outperforms S3D and the LSTM baseline in both representational quality and inference efficiency, achieving higher accuracy at lower latency (1.48 ms/video vs. 2.25 ms/video for S3D). The Transformer-based architecture’s ability to model global temporal dependencies across all 16 frames simultaneously, combined with masked autoencoder pretraining that encourages rich spatiotemporal representations, makes it better suited for capturing the subtle motion dynamics of hand signs.

These findings motivate the decision to proceed with full fine-tuning of the SSv2-pretrained VideoMAE model. We select VideoMAE pretrained on SSv2 as the backbone for fine-tuning, as it demonstrates the strongest frozen feature quality, the best inference efficiency, and the most favorable pretraining domain alignment among all candidates evaluated.

2.2. Fine-Tuning

2.2.1. Full Fine-Tuning

	train_accuracy	test_accuracy	Latency(ms)
S3D	—	—	—
VAE	76.60%	22.08%	1.52

Table 5. FFT on Different Models

We evaluated two models under full fine-tuning: S3D and a VAE-based architecture. S3D is excluded from analysis from the conclusion and experiment we get from Section 2.1. The VAE model achieved a training accuracy of 76.60% but dropped sharply to 22.08% on the test set, indicating significant overfitting. Despite this, the model demonstrated strong inference efficiency, achieving 656.9 FPS with a latency of 1.52 ms per sample, suggesting it is suitable for real-time deployment if the generalization gap can be addressed.

2.2.2. LoRA

We implemented LoRA with $r=16$, $\text{lora_alpha}=32$, and $\text{dropout}=0.2$ in this section.

	train_accuracy	test_accuracy	Parameters
FFT	76.60%	22.08%	87,047,724
LoRA	77.02%	20.63%	822,060

Table 6. Performance on LoRA

Compared full fine-tuning (FFT) trained 87M parameters and achieved 22.08% test accuracy, while LoRA reduced trainable parameters by 99% to 822K yet maintained comparable training accuracy (77.02%) with a marginal

drop in test accuracy to 20.63%. The negligible performance difference suggests that the model’s generalization bottleneck lies in the data rather than model capacity, and LoRA is the preferred choice given its dramatically lower computational cost and comparable results.

2.3. Skeleton-based Sign Language Recognition

From last section, we can observe that. Thus, we explore skeleton-based approaches as an alternative to video-based methods. Rather than processing raw RGB frames, skeleton-based methods extract structured pose information from each frame and use the resulting keypoint sequences as input to the classifier.

We first attempted to extract hand keypoints from WLASL videos using MediaPipe HandLandmarker, producing a feature vector of 126 dimensions per frame (2 hands $\times$ 21 landmarks $\times$ 3 coordinates), sampled uniformly to 32 frames per video. These were stored as .npy files and used to train both an LSTM and a SPOTER (Transformer-based) model. However, this approach yielded poor results. We identify the following reasons:

- WLASL-300 dataset contains only approximately 6.3 samples per class on average, which is insufficient for any model to learn generalizable representations, regardless of architecture.
- MediaPipe HandLandmarker operates on individual frames without temporal context, and frequently fails to detect hands in low-quality or fast-motion frames, leading to a high proportion of zero-padded feature vectors.
- Extracted coordinates are absolute pixel positions without normalization relative to hand size or camera distance, introducing large intra-class variance.

	train_accuracy	test_accuracy	Parameters
SPOTER	0.089	0.06	3,274,028
LSTM	0.568	0.124	2,256,172

Table 7. Performance after LoRA with mediapipe

As a result, both the LSTM and SPOTER models failed to converge meaningfully on this representation. The SPOTER model achieved a validation accuracy of approximately 2–3% after 22 epochs, only marginally above the random baseline of 0.33% for 300 classes. We are currently investigating the use of the official WLASL pre-computed OpenPose keypoints, which may yield better results due to higher detection quality and the inclusion of full-body pose context.

3. Translation System

Given the challenges observed in Section 2, we decided to introduce an alphabet-level recognition module using the Sign Language MNIST dataset first to ensure a fully functional end-to-end pipeline can be demonstrated. The Sign Language MNIST dataset contains 28×28 grayscale images of 24 static ASL hand gestures (excluding J and Z, which require motion), with approximately 27,455 training and 7,172 test samples.

If a suitable replacement for the WLASL keypoint data is identified, we will extend the pipeline to support word-level recognition and update the result in Table 7

3.1. Alphabet-Level Classifier

	Accuracy	Parameters
AlexNet	0.96	23,525,080
ResNet-18	0.99	11,179,992
ResNet-34	0.99	21,288,152
SimpleResNet*	0.99	677,465

Table 8. Benchmark on Different Models

We trained a SimpleResNet on Sign Language MNIST as validation in Section 1, the lightweight model achieved 99.69% test accuracy after only 2 epochs of training, demonstrating that static alphabet recognition is highly tractable with even a small model.

After doing benchmarks on different models in Table 7, as a trade-off between robustness and computing time, we decide use transfer learning on ResNet-18 as our final classifier.

3.2. Alphabet-Level Translator

We first trained a ResNet-18 model on the Sign MNIST dataset. The architecture was adapted for 28×28 single-channel input by replacing the original 7×7 stride-2 conv1 with a 3×3 stride-1 convolution and removing the maxpool layer to preserve spatial information at this resolution. The model was trained from scratch for 10 epochs using Adam optimizer with cosine annealing, achieving over 99% accuracy on the Sign MNIST test set.

However, real-world webcam performance was significantly worse. The core reason is **domain gap**. Sign MNIST images are 28×28 grayscale crops with near-black uniform backgrounds, captured under controlled conditions. Webcam frames contain cluttered backgrounds, varying lighting, different skin tones, and arbitrary hand positions in the frame. The model learned to rely heavily on background and texture cues that simply do not exist in real footage, causing predictions to be unstable and frequently incorrect.

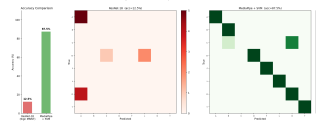


Figure 3. Domain Gap on Sign MNIST

From Figure 3, we can find even with 99% test accuracy on ResNet-18, the actual prediction accuracy through Webcam is only **12.5%**. To address the domain gap, we switched to a representation that is inherently invariant to background and lighting, hand keypoints.

- MediaPipe Hands detects the hand in each webcam frame and extracts 21 landmarks, each with (x, y, z) coordinates — 63 values total.
- The landmarks are normalized: the wrist (point 0) is used as the origin, and distances are scaled by the wrist-to-middle-knuckle (point 9) length. This removes variation due to hand position and distance from the camera.
- A Support Vector Machine (SVM) with RBF kernel is trained on keypoint vectors collected directly from the webcam, one per frame, around 100 samples per letter.



Figure 4. Mediapipe Prediction Examples

3.3. Letter-to-Word Conversion

To move beyond single-letter prediction, we implemented a word-building mechanism after Section 3.2. **Demo is on https://github.com/hd322/CS444.Spring_26.**

We first deploy a stability counter: a letter is only appended to the current word once it has been predicted consistently for $STABLE_FRAMES = 10$ consecutive frames. This filters out transient predictions during hand transitions.

To mark word boundaries, we added a dedicated **SPACE** gesture, an open five-fingered hand, this gesture is treated as its own class in the SVM, trained on webcam keypoint samples like any other letter. When the model detects the **SPACE** gesture, the current accumulated word is committed to the word history and the buffer resets. Besides, a **SPACE_DONE** flag prevents a single held gesture from triggering multiple word commits.

3.4. Word-Level Classifier

In this section, to achieve better performance on WLASL, we introduce I3D [2](Inflated 3D ConvNets). The core idea is to inflate all convolutional filters of a 2D network pretrained on ImageNet from $N \times N$ into $N \times N \times N$, enabling the model to process both spatial and temporal dimensions simultaneously, capturing hand shape and motion trajectory in a single forward pass. The model is further pretrained on the large-scale Kinetics action recognition dataset, giving it strong priors on human body movement before any task-specific fine-tuning.

The baseline approach feeds the full video clip into the pretrained I3D network and uses the resulting 1024-dim feature vector directly for classification via a linear head. While the 3D convolutions capture local spatiotemporal patterns, the model lacks a dedicated mechanism to reason about long-range dependencies between distant frames in a signing sequence.

To address this limitation, we draw inspiration from Video Transformer Network(VTN) [9], which proposes a general paradigm of using a CNN backbone for per-clip feature extraction followed by a Transformer encoder for temporal modeling. VTN demonstrates that decoupling spatial feature extraction from temporal reasoning leads to more effective video understanding, particularly when the temporal structure of the action is complex.

	WLASL-100	WLASL-300	WLASL-1000
I3D	65.89%	56.14%	47.33%
VTN	74.47%	57.86%	45.13%

Table 9. Benchmark on I3D

We follow the original paper, using a 3D-CNN(I3D) backbone for spatiotemporal feature extraction followed by a Transformer encoder(standard 6-layer) for temporal modeling, forming an I3D + Transformer pipeline.

From Table 8 above, we can observe that in WLASL-100/300, since there are fewer classes mean more samples per class. The Transformer has sufficient data to learn meaningful attention patterns across frames, and the signal-to-noise ratio is high enough for the architecture to shine, we can find the performance on VTN is better than I3D only.

However, as the number of classes grows, average samples per class drops sharply in WLASL-1000, the Transformer additional parameters cannot be adequately trained, leading to overfitting, the simpler baseline actually generalizes better under data-sparse conditions.

3.5. Future Improvements

First, in original WLASL [8] paper, the dataset JSON already includes per-sample YOLOv3 bounding boxes, cropping to the signer region before feeding into the model might immediately reduce background noise without any additional annotation effort.

Besides, fusing skeleton keypoints as we introduced in Section 2.3, or replacing I3D with newer video Transformers such as VideoMAE or TimeSformer, or apply heavier data augmentation might also lead to an improvement on the classification.

4. Results

4.1. Experimental Summary

Throughout this project, we evaluated our methodology across four benchmarks: **CIFAR-10** for architectural baselines, **Sign Language MNIST** for baseline and static alphabet classification, **WLASL-100/300/1000** for dynamic word-level recognition, and a custom **Webcam-Collected Dataset** for real-world validation. Moreover, our implementation leverages ViT and Transformer modules adapted from [Assignment 5](#), and [Sanford CS231n](#), alongside pretrained backbones (VideoMAE, I3D) sourced from HuggingFace and official repositories.

4.2. Quantitative Performance

The core metrics from our experiments (detailed in Tables 1–8) are summarized here to highlight key performance trade-offs:

- **CNN vs. ViT (Inductive Bias):** On Sign MNIST, a shallow ResNet-18 (677K parameters) achieved 99.7% accuracy, significantly outperforming a ViT trained from scratch (93.9% with 1.59M parameters), confirming the value of spatial priors in low-data regimes.
- **Transfer Learning Efficiency:** Switching VideoMAE’s pretraining source from Kinetics-400 to SSv2 yielded a **137% relative improvement** (8.5% → 20.2%) in linear probe accuracy. Furthermore, LoRA fine-tuning matched full fine-tuning performance (20.6% vs. 22.1%) while reducing trainable parameters by 99.1%.
- **Temporal Modeling:** VTN (I3D + Transformer) consistently outperformed vanilla I3D on WLASL-100 (74.5% vs. 65.9%) and WLASL-300 (57.9% vs. 56.1%), though this advantage inverted as data sparsity increased in WLASL-1000.
- **The Domain Gap:** Despite achieving 99%+ on Sign MNIST, the ResNet-18 model’s accuracy collapsed to **12.5%** under live webcam conditions. Our proposed MediaPipe + SVM pipeline successfully mitigated this gap.

4.3. Qualitative Analysis

Example predictions and confusion matrices for the Sign MNIST classifier are provided in Figure 1. Figure 4 shows cases MediaPipe’s robust landmark extraction under cluttered background conditions. A live demonstration demo of the end-to-end letter-to-word translation pipeline (incorporating the stability buffer and SPACE gesture) is available at: https://github.com/hd322/CS444_Spring_26.

5. Discussion and Conclusions

This project transitioned from controlled benchmark classification to the deployment of a robust, real-world ASL translation system. Our findings highlight several critical constraints in sign language in the area of CV and AI.

Inductive Biases and Data Scale. Our results confirm that for small-scale sign datasets, the spatial locality and translation-equivariance of CNNs remain superior to the global receptive field of ViTs. Without massive pretraining, the high parameter overhead of Transformers leads to rapid overfitting rather than superior feature extraction.

Pretraining Domain Alignment. The performance leap observed when switching to SSv2 pretraining underscores that *what* a model sees during pretraining is as critical as its architecture. SSv2 is focus on hand-centric temporal dynamics aligns far better with the semantics of ASL than the coarse, full-body actions in Kinetics-400.

Bridging the Domain Gap. The 86.5% performance drop moving from Sign MNIST to webcam input serves as a cautionary tale of “benchmark overfitting”. By pivoting to a MediaPipe-based keypoint representation, we successfully decoupled gesture semantics from environmental noise (lighting, background), providing a blueprint for deployable assistive technologies.

The Temporal Modeling Trade-off. While Transformer-based temporal heads (VTN) improve accuracy on moderate-sized datasets, they are highly sensitive to data density. In the sparse regime of WLASL-1000, simpler spatiotemporal CNNs (I3D) generalize better. This suggests that future ASL research should prioritize data density and augmentation over further architectural scaling.

Summary. We presented a two-stage ASL recognition pipeline that balances real-time efficiency with representational depth. Key findings include: (1) domain-aligned pretraining is the dominant factor in transfer success; (2)

parameter-efficient fine-tuning (LoRA) is ideal for data-scarce sign language tasks; and (3) keypoint-based representations are essential for overcoming the domain gap in real-world deployment.

6. Key Resources

- **Datasets:**
 - [Sign Language MNIST](#)
 - `CIFAR-10`, `torchvision.datasets`
 - [WLASL-300](#)
- **Models & Libraries:**
 - `VideoMAE`, `transformer`
 - `torch`, `opencv-python`, `random`, `peft`, `timm`, `mediapipe`
- **Computing Resources:**
 - Local GPU for real-time inference in Section 3 and validation in Section 1.
 - ICNR for all model fine-tuning did in Section 2.

7. Member Roles

Liuyuan Zhi: Section 1.1 and dataset preparation

Baige He: Linear probe, FFT in Section 2

Ziyi Han: Section 2.2, Section 2.3, and Section 3.4

Hao Dong: Section 1.2 and 1.3, Section 2.2, and Section 3

8. AI Use Plan

- All core computation and network construction till now completed manually.
- AI agent (e.g. Claude, DeepSeek, ChatGPT etc.) used for debugging and idea improving.

9. At the End

Thank you for your tutorial on Deep Learning and Computer Vision! Enjoy your summer break!

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